Meta model that learns across tasks

$$W_{k+1} = W_k - \alpha_k \frac{\partial \mathcal{L}}{\partial W}$$

$W \rightarrow$

$\leftarrow \frac{\partial X}{\partial W}$

$$X_k^*(W_k) = \arg\min_{\bar{X} \in \mathbb{S}^n} f_0(\bar{X}) + g(\bar{X}; W_k)$$
$$\text{subject to } f_i(\bar{X}) \leq a_i, \quad i = 1, \ldots, p,$$
$$h_j(\bar{X}) = b_j, \quad j = 1, \ldots, q,$$
$$X \succeq 0.$$

$X^* \rightarrow$

$\leftarrow \frac{\partial \mathcal{L}}{\partial X}$

$$\mathcal{L}(X_k^*(W_k), W_k)$$

Penalty parameters initialization $W_{1,t+1}$ Predicted by the meta model

Learned rank-1 penalty parameter for task t $W_{K,t}$ Handed to the meta model

ONLINE LEARNING ACROSS TASKS

Meta-model updates parameter from each finished task and predicts $W_{1,t+1}$ for the next one - so fewer inner steps are needed as tasks accumulate